

Programming Fundamental

L – 01

Session 02

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The Computer Programming

- <https://pythonnumericalmethods.berkeley.edu/notebooks/Index.html>

Software Categories

- System SW
 - Programs written for computer systems
 - Compilers, operating systems, ...
- Application SW
 - Programs written for computer users
 - Word-processors, spreadsheets, & other application packages

A Layered View of the Computer

Application Programs

Word-Processors, Spreadsheets,
Database Software, IDEs,
etc...

System Software

Compilers, Interpreters, Preprocessors, etc.
Operating System, Device Drivers

Machine with all its hardware

Operating System (OS)

- Provides several essential services:
 - Loading & running application programs
 - Allocating memory & processor time
 - Providing input & output facilities
 - Managing files of information

Programs

- Programs are written in programming languages
 - PL = programming language
 - Pieces of the same program can be written in different PLs
 - Languages closer to the machine can be more efficient
 - As long as they agree on how to communicate
- A PL is
 - A special purpose and limited language
 - A set of rules and symbols used to construct a computer program
 - A language used to interact with the computer

Computer programming

- Computer Programming is defined as the process of creating computer software using a programming Language.
- Computer programs are written by Human individuals(Programmers)
- A computer program is a step by step set of instructions that a computer has to work through in a logical sequence in order to carry out a particular task.
- The computer executes these instructions (obeys the instructions) when told to do so by the user.

Programming Languages

- Programming languages are the vocabulary and set of grammatical rules for instructing a computer to perform specific tasks.
- There are many different types of programming languages each having a unique set of keywords (words that it understands) and a special syntax (grammar) for organising program instructions.

Syntax

- Syntax refers to the spelling and grammar of a programming language.
- Each program defines its own syntactical rules that control which words the computer understands, which combinations of words are meaningful, and what punctuation is necessary.
- Text-based programming languages are based on sequences of characters, while visual programming languages are based on the spatial layout and connections between symbols (which may be textual or graphical).

Learn A Language

- There are many computer programming languages and so the programmer will have to decide which one to use for solving a particular problem. These languages must be learnt just as Swahili, English or French etc.

The generation process of code

**The roles of the pre-processor, compiler and
linker, interpreter**

Levels of programming languages

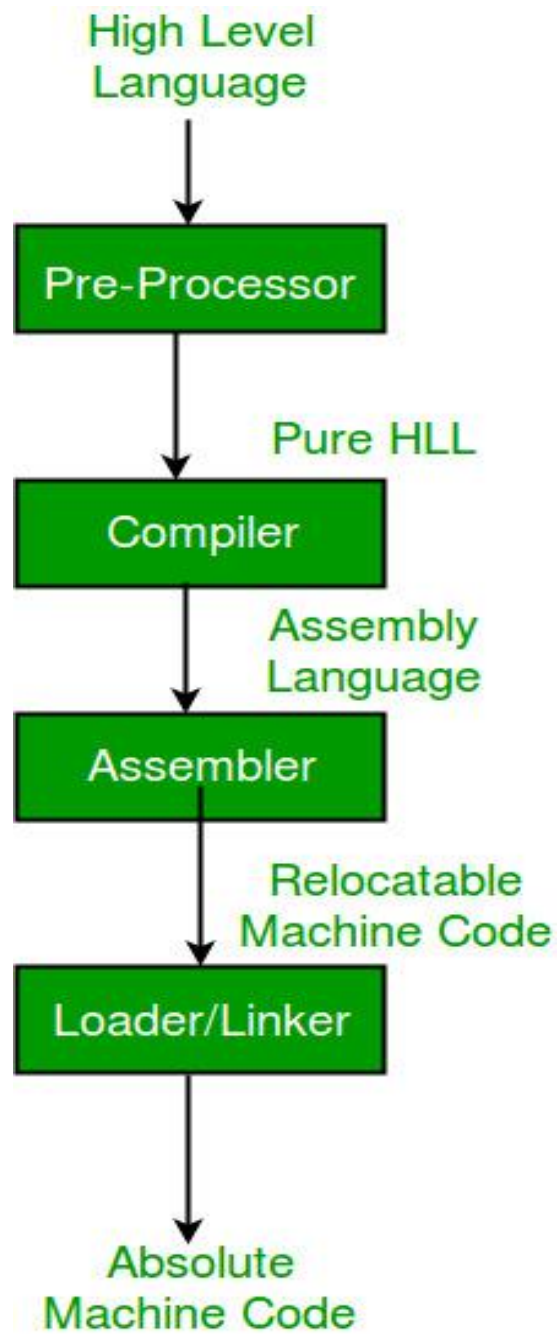
- Programming languages are classified into five major categories:
 - machine languages (first generation languages),
 - assembly languages (second generation languages),
 - third generation languages,
 - fourth generation languages, and
 - natural languages.
- Machine and assembly languages are referred to as low level languages; third generation, fourth generation and natural languages are categorised as high level languages.

High-level programming languages

- High level programming language is defined as one that is machine independent and uses:
 - variables and objects
 - Boolean expressions
 - functions
 - Loops
 - Arrays
 - Threads
 - locks which are similar to their meaning (abstraction).
- High-level languages have evolved over the years and can be grouped into five categories:
 - Third Generation Languages (3GL)
 - Fourth Generation Languages (4GL)
 - Object Oriented Programming Languages (OOP)
 - Fifth Generation Languages (5GL) and
 - Scripting Languages.

Language Processing Systems

- We know a computer is a logical assembly of Software and Hardware. The hardware knows a language, that is hard for us to grasp, consequently, we tend to write programs in a high-level language, that is much less complicated for us to comprehend and maintain in our thoughts.
- Now, these programs go through a series of transformations so that they can readily be used by machines. This is where language procedure systems come in handy.



Pre-Processor

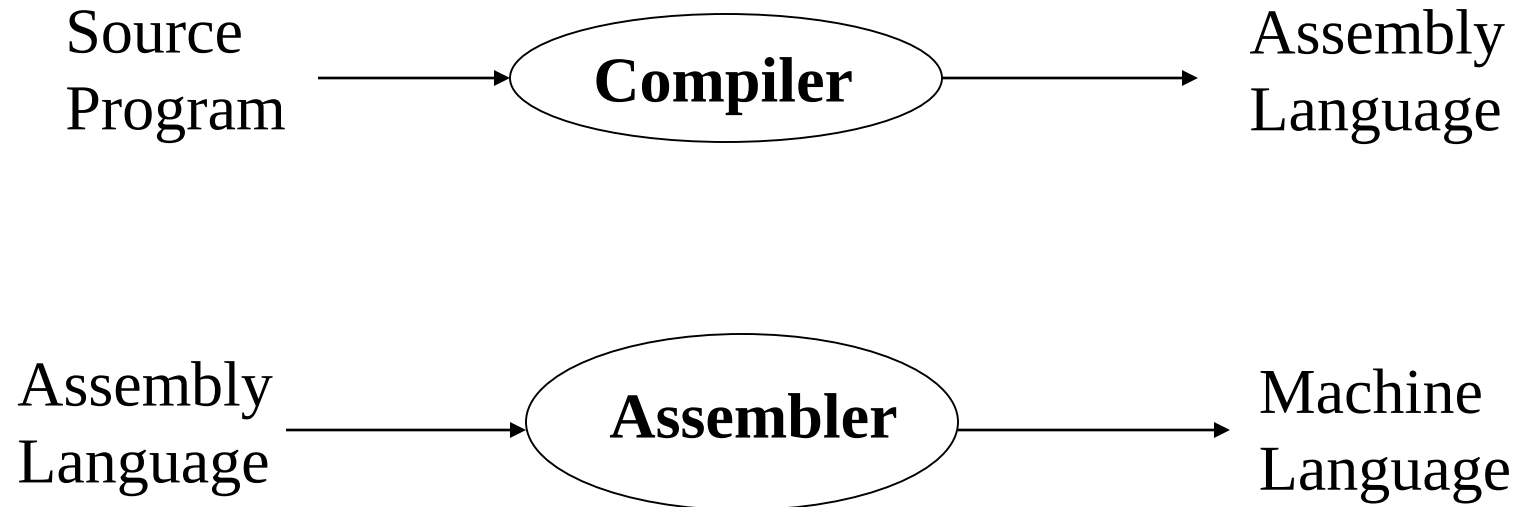
- Pre-processing refers to the transformations applied to your data before feeding it to the algorithm
- However in Python, a preprocessor is not as explicit or commonly used as in some other programming languages like C or C++. Python is a dynamically typed and interpreted language, and it doesn't have a separate compilation step where a preprocessor might be employed

Compilers & Programs

- **Compiler**

- A program that converts another program from some source language (or high-level programming language / HLL) to machine language (object code).
- Some compilers output assembly language which is then converted to machine language by a separate assembler.
- Is distinguished from an assembler by the fact that each input statement, in general, correspond to more than one machine instruction.

Compilation into Assembly L



Assembly vs. machine code

Machine code bytes	Assembly language statements
	foo:
B8 22 11 00 FF	movl \$0xFF001122, %eax
01 CA	addl %ecx, %edx
31 F6	xorl %esi, %esi
53	pushl %ebx
8B 5C 24 04	movl 4(%esp), %ebx
8D 34 48	leal (%eax,%ecx,2), %esi
39 C3	cmpl %eax, %ebx
72 EB	jnae foo
C3	retl

Instruction stream

B8 22 11 00 FF 01 CA 31 F6 53 8B 5C 24
04 8D 34 48 39 C3 72 EB C3

Compilers & Programs

- **Source program**

- The form in which a computer program, written in some formal programming language, is written by the programmer.
- Can be compiled automatically into object code or machine code or executed by an interpreter.
- Python source programs have extension '.py'

Compilers & Programs

- **Object program**

- Output from the compiler
- Equivalent machine language translation of the source program
- Files usually have extension '.obj'

- **Executable program**

- Output from linker/loader
- Machine language program linked with necessary libraries & other files
- Files usually have extension '.exe'

Language Translators

- Language translators are system programs that convert assembly language and high-level language into the machine language.
- The computer does not understand assembly languages or high-level languages.
- Computers work in machine code or machine language.

Source code

- Source code is a program code of either an assembly language or high-level language that the programmers write in a program editor.
- Source code cannot be understood by the computer until it has been translated into machine code.

Object code

- Object code is a program code in machine-readable form (a source program that has been translated).

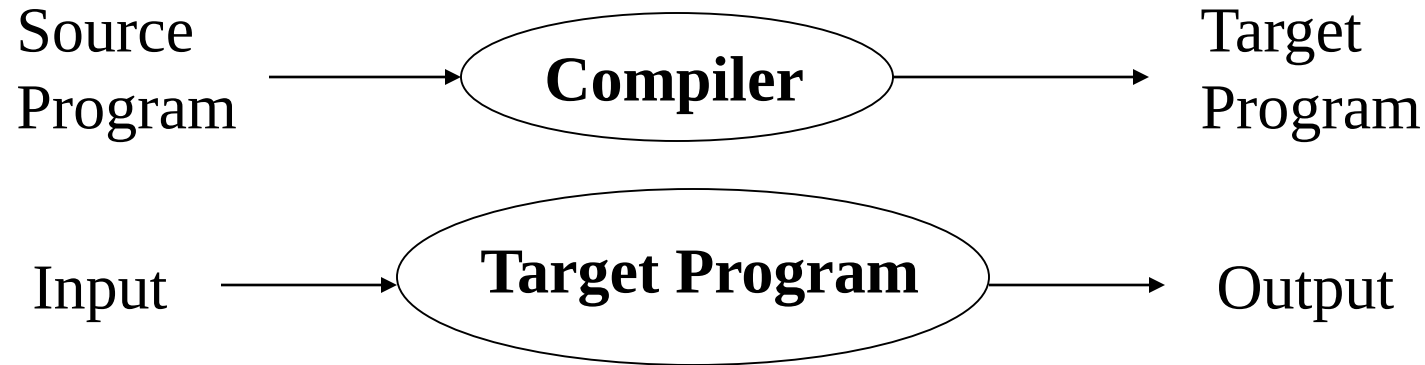
Assemblers

- The assembly-language program must be translated into machine code by a separate program called an assembler.
- The assembler program recognises the character strings that make up the symbolic names of the various machine operations, and substitutes the required machine code for each instruction.

Compilers

- Compilers are language translators that translate high-level languages into machine code.
- A compiler translates the entire program (source code) to machine code, and then the code is executed.
- The translated codes are known as object codes and are saved as a file on disk.
- The object code (machine code) stored on disk has an EXE file name extension. It is then loaded or linked (stored in memory) and executed

Compilation

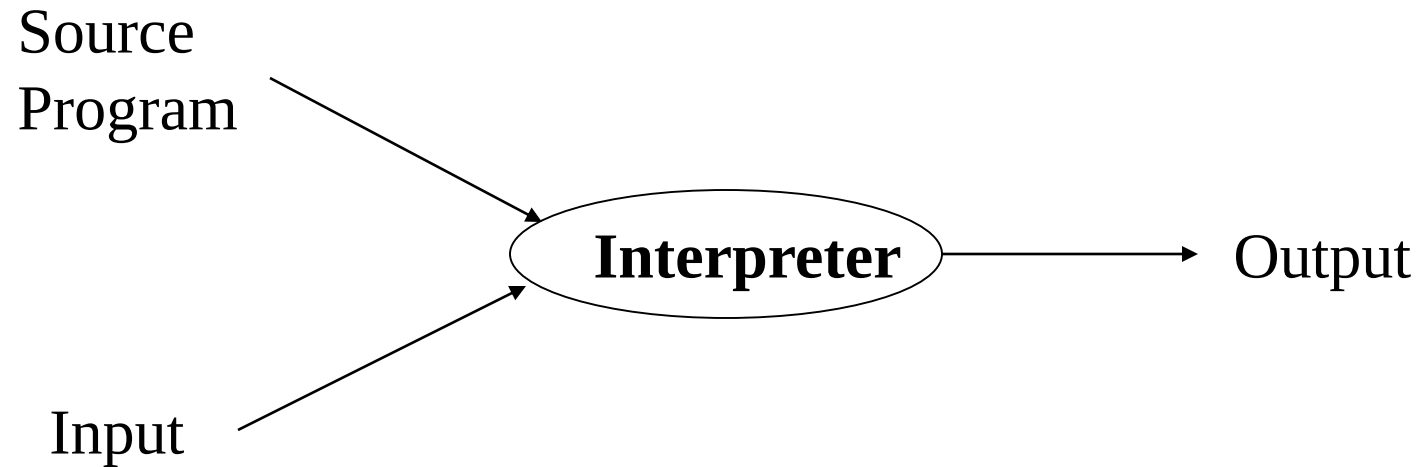


- Compiler translates source into target (a machine language program)
- Compiler goes away at execution time
- Compiler is itself a machine language program, presumably created by compiling some other high-level program
- Machine language, when written in a format understood by the OS is **object code**

Interpreters

- Interpreters are language translators that translate high-level language into machine code.
- An interpreter translates and executes one instruction at a time as it is encountered.
- The machine codes are not saved after execution

Interpretation



- The interpreter stays around during execution
- It reads and executes statements one at a time

Compilation vs. Interpretation

- Compilation:
 - Syntax errors caught before running the program
 - Better performance
 - Decisions made once, at compile time
- Interpretation:
 - Better diagnostics (error messages)
 - More flexibility
 - Supports **late binding** (delaying decisions about program implementation until runtime)
 - Can better cope with PLs where type and size of variables depend on input
 - Supports creation/modification of program code on the fly (e.g. Python, Lisp, Prolog)

Loader/Linker

- It converts the reloadable code into absolute code and tries to run the program resulting in a running program or an error message (or sometimes both can happen).
- Linker loads a variety of object files into a single file to make it executable.
- Then loader loads it in memory and executes it.

Program Errors

- Syntax Errors:
 - Errors in grammar of the language
- Runtime error:
 - When there are no syntax errors, but the program can't complete execution
 - Divide by zero
 - Invalid input data
- Logical errors:
 - The program completes execution, but delivers incorrect results
 - Incorrect usage of parentheses

Python Programming Language

- Python is a very popular general-purpose interpreted, interactive, object-oriented, and high-level programming language.
- Python is dynamically-typed and garbage-collected programming language.
- It was created by Guido van Rossum during 1985- 1990.
- Python source code is also available under the GNU General Public License (GPL).

- Python is currently the most widely used multi-purpose, high-level programming language, which allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and the indentation requirement of the language, makes them readable all the time.

Why Python?

- Python is Open Source which means its available free of cost.
- Python is simple and so easy to learn
- Python is versatile and can be used to create many different things.
- Python has powerful development libraries include AI, ML etc.
- Python is much in demand and ensures high salary

Python Advantages

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- Python is Object-Oriented – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- Python is a Beginner's Language – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

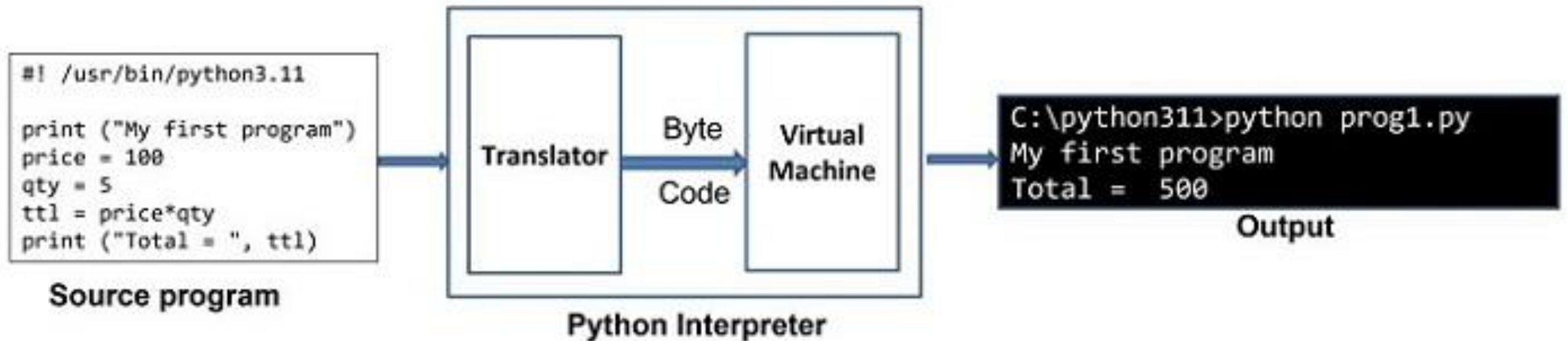
If you are a Python developer, you are in high demand. There are many jobs available in the current IT/Software development market for fresher who know Python well.

- Web developer
- Backend developer
- Data scientist
- Machine learning engineer
- Software engineer
- Systems Engineer
- DevOps Engineer
- Cloud engineer
- Mobile Developer
- Game developer
- Research engineer
- Quality assurance engineer
- Technical writer
- Product Manager
- Technical support specialist

Python - Hello World Program

- `Print("Hello World!")`

Python Interpreter



Python Reserved Words

and	as	assert
break	class	continue
def	del	elif
else	except	False
finally	for	from
global	if	import
in	is	lambda
None	nonlocal	not
or	pass	raise
return	True	try
while	with	yield

Python Variables - Naming Convention

- Name of Python variable should start with either an alphabet (lower or upper case) or underscore (_). More than one alpha-numeric characters or underscores may follow.
- Use of any keyword as Python variable is not allowed, as keywords have a predefined meaning.
- Name of a variable in Python is case sensitive. As a result, age and Age cannot be used interchangeably.
- You should choose the name of variable that is mnemonic, such that it indicates the purpose. It should not be very short, but not very lengthy either.

Python Variables - Naming Convention

- Camel case – First letter is a lowercase, but first letter of each subsequent word is in uppercase. For example: kmPerHour, pricePerLitre
- Pascal case – First letter of each word is in uppercase. For example: KmPerHour, PricePerLitre
- Snake case – Use single underscore (_) character to separate words. For example: km_per_hour, price_per_litre

Python - Comments

- Python comments are programmer-readable explanation or annotations in the Python source code.
- They are added with the purpose of making the source code easier for humans to understand, and are ignored by Python interpreter.
- Comments enhance the readability of the code and help the programmers to understand the code very carefully.
- There are three types of comments available in Python
 - Single line Comments
 - Multiline Comments
 - Docstring Comments

Comments

```
# This is a single line comment in python  
  
print ("Hello, World!")
```

```
# This is a comment.  
# This is a comment, too.  
# This is a comment, too.  
# I said that already.
```

```
...  
This is a multiline  
comment.  
...
```

Python - Data Types

- Numeric - int, float, complex
- String - str
- Sequence - list, tuple, range
- Binary - bytes, bytearray, memoryview
- Mapping - dict
- Boolean - bool
- Set - set, frozenset
- None - NoneType

Python Numeric Data Type

int	long	float	complex
10	51924361L	0.0	3.14j
100	-0x19323L	15.20	45.j
-786	0122L	-21.9	9.322e-36j
080	0xDEFABCEC BDAECBFBAE l	32.3+e18	.876j
-0490	5356336298 43L	-90.	-.6545+0J
-0x260	- 0523181727 35L	-32.54e100	3e+26J
0x69	- 4721885298 529L	70.2-E12	4.53e-7j

Python String Data Type

- Python Strings are identified as a contiguous set of characters represented in the quotation marks.
- Python allows for either pairs of single or double quotes.
- Subsets of strings can be taken using the slice operator (`[ ]` and `[:]`) with indexes starting at 0 in the beginning of the string and working their way from -1 at the end.

Practice

```
str = 'Hello World!'
```

Edit & Run 

```
print (str)           # Prints complete string
print (str[0])         # Prints first character of the string
print (str[2:5])       # Prints characters starting from 3rd to 5th
print (str[2:])        # Prints string starting from 3rd character
print (str * 2)        # Prints string two times
print (str + "TEST")  # Prints concatenated string
```



Python List Data Type

- A Python list contains items separated by commas and enclosed within square brackets ([]).
- To some extent, Python lists are similar to arrays in C.
- One difference between them is that all the items belonging to a Python list can be of different data type where as C array can store elements related to a particular data type.

Practice

```
list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]  
tinylist = [123, 'john']
```

[Edit & Run](#) 

```
print (list)           # Prints complete list  
print (list[0])        # Prints first element of the list  
print (list[1:3])      # Prints elements starting from 2nd till 3rd  
print (list[2:])        # Prints elements starting from 3rd element  
print (tinylist * 2)    # Prints list two times  
print (list + tinylist) # Prints concatenated lists
```

Python Tuple Data Type

- A Python tuple consists of a number of values separated by commas. Unlike lists, however, tuples are enclosed within parentheses.

```
my_tuple = (1, 2, 3, 'apple', 'banana')
```

```
# Accessing elements
```

```
print(my_tuple[0])      # Output: 1
```

```
print(my_tuple[3])      # Output: 'apple'
```

```
# Slicing
```

```
print(my_tuple[1:4])    # Output: (2, 3, 'apple')
```

```
# Nested tuples
```

```
nested_tuple = (1, 2, ('apple', 'banana'), 3)
```

```
print(nested_tuple[2])  # Output: ('apple', 'banana')
```

```
print(nested_tuple[2][0])# Output: 'apple'
```

Python Lines and Indentation